Grade 6 General Science

Solved Exercises - Balochistan Textbook Board

Balochistan Textbook Board (BTBB)

Unit 1: Cellular Organization

Exercise 1.1: Tick the correct option (pages 32-33)

1. The basic unit of structure and function of living things is:

- ✓ Cell
- Tissue
- Organ
- Organ system

2. Which of the following is NOT present in an animal cell?

- Cytoplasm
- Nucleus
- ✓ Cell wall
- Vacuole

3. The outer most covering of a plant cell is:

- ✓ Cell wall
- Cell membrane
- Cytoplasm
- Chloroplast

4. The chromosomes are found in:

- Cytoplasm
- ✓ Nucleus
- Vacuole
- Lysosome

5. The power house of cell is:

- Nucleus
- Plasma membrane
- Cytoplasm
- ✓ Mitochondria

6. Plant cells contain:

- ✓ A single large vacuole
- Many large vacuoles
- Many small vacuole
- One small vacuole

7. The animal cell lacks:

- Cell membrane
- Mitochondria
- Endoplasmic reticulum
- ✓ Chloroplast

8. A group of similar cells together make:

- Organism
- ✓ Tissue
- Organ
- Organ system

9. The oxygen in the body is carried by the:

- Muscle cells
- White blood cells

✓ Red blood cells

Platlets

10. The transport of water from roots to leaves occurs through:

✓ Xylem

Phloem

Epidermis

Mesophyll

Exercise 1.2: Write short answers (page 33)

1. Define cell.

A cell is the basic unit of structure and function of living things. Living things are made up of cells; cells are very small structures which cannot be seen with the naked eye and are studied with the help of a microscope. The literal meaning of 'cell' is a small room or a bounded space.

2. Who introduced the term cell?

The term 'cell' was introduced in 1665 by the British scientist Robert Hooke. He examined a thin slice of cork under a microscope and observed honeycomb-like structures, which he called cells.

3. Name the cell organelle in which green pigment chlorophyll is present?

Chloroplast. It contains the green pigment called chlorophyll, which is why plant leaves look green, and it helps plants prepare their own food by photosynthesis. Chloroplasts are present in plant cells and absent in animal cells.

4. How the position of vacuole in a plant cell differs from the animal cell?

In a plant cell there is a single large vacuole in the center, which pushes the nucleus towards the cell membrane. In an animal cell there are many small vacuoles, and the nucleus lies almost in the center of the cell. A vacuole is a sac-like structure that stores salts, minerals and proteins and isolates waste products that might be harmful to the cell.

5. What function is performed by the white blood cells?

White blood cells are part of our immune system: they protect us from diseases because they fight against germs and kill them. They lack haemoglobin and contain a nucleus, they are irregular in shape, and they are generally the largest cell type found in the blood.

Exercise 1.3: Write detailed answers (page 33)

1. Compare and contrast between plant and an animal cell on the basis of cell organelles.

Similarities: both plant and animal cells have a cell membrane, cytoplasm, nucleus, mitochondria, ribosomes, endoplasmic reticulum, Golgi apparatus, lysosomes and vacuoles. Differences: a plant cell has a cell wall (outermost covering, mainly cellulose, hard, protective) while an animal cell has none; a plant cell has chloroplasts (with chlorophyll, for photosynthesis) while an animal cell has none; a plant cell has a single large central vacuole while an animal cell has many small vacuoles; in a plant cell the large

vacuole pushes the nucleus towards the cell membrane, while in an animal cell the nucleus is almost central; centrioles are present near the nucleus of the animal cell and absent in plant cells.

2. Make a list of plant organs. How these organs are organized into root and shoot system?

Plant organs are root, stem, leaves, flower and fruit. They are organized into two systems. Root system: the root and its branches, which absorb water and minerals from the soil. Shoot system: the stem, its branches, leaves and flowers — the above-ground part of the plant, which provides support to the whole structure. The vascular system (xylem and phloem) transports water and food to different parts of the plant, and the flower is the part of the reproductive system which produces seeds and fruits.

3. Make a list of human organs. Explain how these organs are arranged into different organ systems?

Human organs include the heart, lungs, kidneys, liver, eyes, brain, stomach, small intestine and large intestine. A group of organs that work together to perform a specific function is called an organ system. For example, the digestive system (mouth, stomach, intestines, liver) digests food; the respiratory system (lungs and air passages) takes in oxygen and removes carbon dioxide; the blood circulatory system (heart and blood vessels) transports blood; and the nervous system (brain, nerves) controls movement, sensation and thinking. The human body has 11 major organ systems, and different organ systems work together to keep the body alive.

4. Describe the types of cells and their functions which are present in blood.

Blood is a liquid tissue consisting of blood cells in plasma. Its three types of cells are: (1) Red blood cells: disc-shaped, they circulate in the body and deliver oxygen to the cells and remove carbon dioxide from the body; they contain haemoglobin, which helps transport oxygen and remove carbon dioxide through the lungs, and they live about 120 days. (2) White blood cells: irregular in shape, lacking haemoglobin and containing a nucleus; they are part of the immune system and protect us from diseases by fighting and killing germs. (3) Platelets: the smallest blood cells, looking like tiny plates under the microscope, without a nucleus; they help the body form clots to stop bleeding. Plasma is the liquid content of blood tissue, which distributes nutrients, removes wastes and prevents infection.

5. Relate the structure of some common cells as Nerve cells, Muscle cells, Epithelium cells and blood cells to their functions.

Nerve cells (neurons): they have long fibers and are branched; they are the basic functional unit of the nervous system and carry messages from the body to the brain and back to the body. Muscle cells: they contract and relax to produce movement. Epithelium cells: they are flat and tightly packed, found in layers covering the surfaces of our skin, the inner lining of blood vessels and the alimentary canal; they perform protection, secretion and diffusion, and protect us from infections. Blood cells: red blood cells carry oxygen and remove carbon dioxide, white blood cells fight germs as part of the immune system, and platelets form clots to stop bleeding. In each case the shape fits the function: long branched

nerve cells carry messages over distances, contracting muscle cells move parts, flat tight epithelial cells form protective coverings, and disc-shaped red cells move easily through vessels.

Exercise 1.4: Match column A with column B (page 34)

Cell wall is composed of -> ?

Cell wall is composed of Cellulose. (The cell wall is the outermost covering of a plant cell, mainly made up of cellulose; it is hard and protects the plant cell.)

Nucleus -> ?

Nucleus -> Controls all functions of the cell. (The nucleus is the most prominent part of a cell; chromosomes, made up of DNA and protein, are present in it.)

Chloroplast contains -> ?

Chloroplast contains Chlorophyll. (Chloroplast is present in plant cells, absent in animal cells; it helps plants prepare food by photosynthesis.)

Cell membrane is composed of -> ?

Cell membrane is composed of Lipids and protein. (The cell membrane controls the movement of materials into and out of the cell; it is a partially permeable membrane.)

Vacuole -> ?

Vacuole -> A sac-like structure. (A single large vacuole is present in the center of a plant cell; animal cells contain many small

vacuoles.)

Chromosomes -> ?

Chromosomes -> Thread-like structures. (Chromosomes are present in the nucleus.)

DNA -> ?

DNA -> Deoxyribonucleic acid.

Chromosomes are made up of -> ?

Chromosomes are made up of DNA and protein. (The extra Column B entry 'DNA and carbohydrate' matches nothing and is left unused.)

Exercise 1.5: Label the following diagrams (page 34)

Label the two cell diagrams.

Left diagram: a plant cell. Label: cell wall (outermost covering), cell membrane (just inside the cell wall), cytoplasm, nucleus (pushed towards the edge by the large central vacuole), chloroplasts (green, with chlorophyll), mitochondria, and the single large central vacuole (sap vacuole). Right diagram: an animal cell. Label: cell membrane (outermost covering), cytoplasm, nucleus (almost central), mitochondria, Golgi apparatus, endoplasmic reticulum, lysosomes, ribosomes and many small vacuoles. Note what is absent from the animal cell: no cell wall, no chloroplasts, no single large vacuole.

Exercise 1.6: Project (page 35)

Project: With the help of your teacher, collect information through

the internet about the structure and function of the following plant and animal cells — plant cells: xylem cells, phloem cells, root hair cells; animal cells: muscle cells, heart cells, nerve cells — and discuss with your classmates how the structure of the cells relates to the function of these cells. Guidance from the unit: xylem cells are long tube-shaped and conduct water from roots to upper parts; phloem cells are tubular and transport food made in the leaves to all other parts; muscle cells contract and relax to produce movement; nerve cells have long branched fibers and carry messages. Root hair cells and heart cells are not described in this unit, so collect their details from the internet with your teacher's help and write them in your own words.

Unit 2: Reproduction in Plants

Exercise 2.1: Tick the correct option (pages 49-50)

1. The process by which living things produce offsprings is called:

- Growth
- Respiration
- Photosynthesis
- ✓ Reproduction

2. The fusion of gametes is involved in:

- Asexual Reproduction
- ✓ Sexual Reproduction
- Vegetative Propagation
- Layering

3. The male part of flower is:

- Sepals
- Petals
- ✓ Stemens
- Carpel

4. Female reproductive part of a flower is:

- Anther
- Filament
- ✓ Carpel
- Sepal

5. Transfer of pollen grains from anther to stigma is called:

- Fertilization
- ✓ Pollination
- Propagation
- Reproduction

6. Pollen grains are received by:

- Filament
- Anther
- ✓ Stigma
- Style

7. Underground stem with thick leaves is known as:

- Runner
- Graft
- ✓ Bulb
- Tuber

8. Potato is an example of:

- Bulb
- Scion
- Stock
- ✓ Tuber

9. The reproductive part of a plant is:

- Roots
- Stem

Leaves

✓ Flower

10. Natural vegetative propagation in plants takes place by:

Flowers

Stamens

Carpel

✓ Bulb

Exercise 2.2: Write short answer (page 50)

1. Define reproduction.

Reproduction is the biological process by which organisms produce offsprings of their own kind. It is the basic property of all living things and ensures the survival of living organisms and the continuation of plant species.

2. Differentiate between sexual and asexual reproduction.

Sexual reproduction involves the fusion of male and female gametes (sperms and eggs); it requires two parents, and the offsprings have characteristics of both parents and are not identical. It typically involves flowers, fruits and seeds in plants. Asexual reproduction involves no fusion of gametes; only one parent is required, the offsprings are identical, and it uses vegetative parts of plants like bulbs, tubers and runners.

3. How self-pollination differs from cross-pollination?

Self-pollination is the transfer of pollen grains from the anther of a flower to the stigma of the same flower, or to the stigma of another

flower on the same plant; it takes place in pea, tomato and cotton plants. Cross-pollination is the transfer of pollen grains from the anther of a flower to the stigma of a flower on another plant of the same species; it takes place in maize, rose and papaya.

4. Write names of four parts of flower.

The four main parts of a flower are sepals, petals, stamens and carpels.

5. What are the vegetative parts of a plant?

The vegetative parts of a plant are roots, stem, leaves and branches. New plants growing from vegetative parts of the parent plant is called vegetative propagation; the flower (with its reproductive parts) is the reproductive part of the plant.

Exercise 2.3: Write detailed answer (pages 50-51)

1. Describe the structure of a flower.

The flower is the reproductive part of a plant, and it consists of four main parts. Sepals: green, leaf-like structures which protect the flower in the bud stage. Petals: coloured, showy, attractive leaf-like structures above the sepals, which attract birds and insects for pollination. Stamens: the male reproductive part, consisting of two parts — the anther (the swollen part, in which pollen grains are formed, producing the male gametes) and the filament (a long stalk-like structure). Carpel: the female reproductive part, consisting of three parts — the stigma (at the top, which receives pollen grains during pollination), the style (a long tube-like structure which passes the pollen grains to the ovary), and the

ovary (the swollen basal part, containing the ovule which bears the female gamete, the egg).

2. Elaborate the process of sexual reproduction in plants.

Sexual reproduction in plants involves the fusion of male and female gametes, and flowering plants reproduce sexually. Its processes are: (1) Pollination: transfer of pollen grains from the anther to the stigma (self-pollination within the same flower or plant; cross-pollination to another plant of the same species). (2) Fertilization: when the pollen tube enters the ovary, it releases its sperms into the ovule; one of the two sperms fuses with the egg and forms a zygote. (3) Seed formation: after fertilization the zygote develops into an embryo, the ovule containing the embryo forms a seed, and the ovary grows large and develops into a fruit, which protects the seeds; the embryo gradually develops into cotyledons, plumule and radicle. (4) Seed dispersal: seeds are dispersed by wind, animals or water. (5) Seed germination: fully mature seeds sown in soil germinate when favourable conditions like moisture and suitable temperature are available, and a seedling emerges which grows into a new plant.

3. How do plants reproduce asexually?

Asexual reproduction in plants (vegetative propagation) is reproduction in which the fusion of male and female gametes does not occur, and new plants grow from vegetative parts of the parent plant. Natural vegetative propagation occurs through underground stems (bulb, tuber) and runners. A bulb is an underground stem with thick leaves (the leaves store food); onion, tulip and daffodil

have bulbs, and bulbs reproduce through buds — a bud is an outgrowth which emerges in the centre of the bulb, and when buried in the soil a new plant grows from it. A tuber is an underground thick stem; potato is an example, producing buds known as 'eyes', and pieces of potato having 'eyes' buried in soil grow into new plants. Runners are special stems which spread over the ground horizontally; grasses and strawberries reproduce through runners, which have nodes where buds are present, and new plants grow from these buds. Artificial vegetative propagation (practised by humans for hundreds of years) includes cutting, grafting, layering and budding.

4. Differentiate between bulb and a tuber.

A bulb is an underground stem with thick leaves; the leaves store food, and it reproduces through buds (onion, tulip, daffodil). A tuber is an underground thick stem which produces buds known as 'eyes' (potato); pieces of potato having 'eyes' buried in the soil grow into new plants. So both are underground stems, but the bulb stores food in thick leaves around a central bud, while the tuber is itself a thickened stem with 'eyes' as buds.

5. What are the advantages of artificial propagation of plants?

The advantages of artificial propagation are: a large number of plants can be grown rapidly with desired traits; plants can be propagated throughout the year; rare and endangered plant species can be protected; yield and quality of crop plants is enhanced; disease and drought resistant plant varieties can be developed; seedless fruits can be produced (for example oranges,

grapes); and good characteristics of two different varieties can be combined in a new plant. Grafting and budding techniques are used for the production of mangoes and many other woody and nursery plants, and they enhance yield and disease-resistance.

6. Distinguish between artificial and natural asexual reproduction in plants.

Natural asexual reproduction happens by itself through underground stems (bulb, tuber) and runners: bulbs reproduce through buds, tubers through 'eyes', and runners through buds at their nodes. Artificial asexual reproduction is practised by humans and includes: cutting (a piece of plant such as a stem having buds is cut and planted in soil; buds on the underground part grow into roots, buds on the aerial part grow into stem and leaves; rose, grapes, sugar cane); grafting (a cut piece of a plant, the scion, is joined with another plant, the stock, so both appear grown as a single plant); layering (a part of a stem is covered or buried in soil without cutting it from the parent plant; roots grow from the covered part, and when the new plant is established it is cut from the parent); and budding (a bud from one plant is grafted onto the stem of another plant).

Exercise 2.4: Match column A with column B (page 51)

Gamete -> ?

Gamete -> Sperms and eggs. (The male gamete is called sperm and the female gamete is known as egg.)

Leaf-like structure that protects flower in bud -> ?

Leaf-like structure that protects flower in bud -> Sepal. (Sepals are green leaf-like structures which protect the flower in the bud stage.)

Special stem that spread over the ground -> ?

Special stem that spread over the ground -> Runner. (Runners are special stems which spread over the ground horizontally; grasses and strawberries reproduce through runners.)

The plant on which scion is attached -> ?

The plant on which scion is attached -> Stock. (In grafting, the scion is joined with the stock plant.)

Production of identical offsprings -> ?

Production of identical offsprings -> Asexual reproduction. (In asexual reproduction no gametes fuse and the offsprings are identical.)

A type of reproduction that required two parents -> ?

A type of reproduction that required two parents -> Sexual reproduction. (Sexual reproduction involves the fusion of male and female gametes.)

Production of fruits without seeds -> ?

Production of fruits without seeds -> Artificial vegetative propagation. (The unit lists 'produce seedless fruits, e.g. oranges, grapes' among the advantages of artificial propagation. The extra Column B entry 'scion' matches nothing and is left unused.)

Exercise 2.5: Label different parts of flower in the following diagram (page 52)

Label the flower diagram.

Label the flower diagram: sepals (green leaf-like structures at the base, protecting the bud), petals (coloured, showy structures above the sepals), stamen (male part: anther, the swollen part producing pollen grains, and filament, the long stalk), carpel (female part: stigma at the top receiving pollen, style, the long tube passing pollen to the ovary, and ovary at the base containing the ovule with the female gamete), plus receptacle (the base on which the flower parts sit) and ovule inside the ovary. (The diagram itself is printed on page 52 of the book; the labels above follow the unit's own flower description, sections 2.1.1 and Figure 2.1.)

Exercise 2.6: Project (page 52)

Project: Investigate the importance of pollinators in the process of sexual reproduction of plants; the effects of pesticides on the populations of pollinators; and what will be the effects on the yield of crop plants if the populations of pollinators are decreased.

Guidance from the unit: petals attract birds and insects for pollination, and pollination (transfer of pollen from anther to stigma) is the first step of sexual reproduction, leading to fertilization, seed and fruit formation. So fewer pollinators means less pollination, fewer seeds and fruits, and lower crop yield. Pesticides can kill or reduce pollinators such as insects and birds, which is why their populations fall. Write your investigation in your own words with examples from your area.

Unit 3: Balanced Diet

Exercise 4.1: Tick the correct option (pages 65-66) [the book numbers this unit's exercises 4.1-4.4, then 3.5-3.6]

1. The body of humans need:

- Less food
- Excessive food
- Food rich in fats
- ✓ Balanced diet

2. Iron deficiency causes:

- Weight loss
- Rickets
- Night blindness
- ✓ Anaemia

3. The primary source of energy for human body is:

- Fats
- Proteins
- ✓ Carbohydrates
- Minerals

4. Source of vitamin A is:

- ✓ Carrot
- Citrus fruits
- Banana

Peas

5. Goiter is caused due to lack of:

Iron

✓ Iodine

Calcium

Magnesium

6. Deficiency of protein causes:

Anaemia

Fever

✓ Kwashiorkor

Night blindness

7. The rich source of vitamin C is?

Red meat

Potato

✓ Citrus fruits

Spinach

8. The deficiency of calcium is characterized by:

Blood clotting

✓ Weak bones

Muscles contraction

Strong teeth

9. Bleeding gums is caused by the deficiency of vitamin:

A

✓ C

D

K

10. The percentage of water in our body is:

40-60%

✓ 55-60%

70-80%

80-90%

Note: the book mislabels the options of Q9 and Q10, printing (b) twice in the second row of each. Read in order, Q9's options are A, C, D, K (correct: C) and Q10's are 40-60%, 55-60%, 70-80%, 80-90% (correct: 55-60%).

Exercise 4.2: Write short answers (page 66)

1. Define balanced diet.

A balanced diet is the diet that contains all the essential nutrients which are required by a person for living a healthy life. It contains all the constituents of food — carbohydrates, proteins, fats, minerals, vitamins, water and fiber — in precise proportions: a good proportion of carbohydrate and protein and a small amount of fats, with the minerals and vitamins essential for health.

2. Differentiate between diet and balanced diet?

Diet is the food and drinks we consume daily. A balanced diet is a diet that contains all the essential nutrients required for living a healthy life, in precise proportions. A diet that lacks the essential components needed by the human body for normal functioning is

known as an unbalanced diet, and it causes nutritional deficiency disorders.

3. What are five main groups of food?

On the basis of nutrients, foods are classified into five major groups: carbohydrates, proteins, fats and oils, vitamins, and minerals. (A balanced diet also includes water and fiber.)

4. Write names of any five carbohydrate rich foods.

Bread, rice, cereals, milk and honey. (Other carbohydrate-rich foods named in the unit: sugar, starch, glucose, fruits and vegetables.)

5. What are proteins? Why do we need them?

Proteins are large molecules made up of amino acids; there are about 20 different types of amino acids. We need them because they are needed to build our body tissues when we are growing, and they help in the building and repair of body tissues. Enzymes and hormones are built up of protein, and proteins are part of antibodies which fight against disease-causing bacteria. Animal sources: meat (chicken, mutton, beef, fish), eggs, milk, cheese, yogurt. Plant sources: beans, pulses, nuts, seeds, peas, legumes, oats. On average, protein makes up about 16% of a person's body weight.

Exercise 4.3: Write detailed answers (page 66)

1. Describe carbohydrates as essential component of diet.

Carbohydrates are the main and primary source of energy for our body — they act as fuel. They include glucose, sugar and starch,

and they are made up of carbon, hydrogen and oxygen. We get them from bread, rice, cereals, milk, honey, fruits and vegetables. They are a quick source of energy: they are absorbed in the wall of the small intestine and reach every cell of our body through blood, and they release energy when broken down by chemical reactions in our body. Carbohydrates are stored in the liver in the form of glycogen. Our brain almost completely depends upon carbohydrates for its energy needs. Carbohydrate deficiency causes fatigue, weakness, low weight and constipation.

2. Write a note on fats explaining its dietary significance.

Fats are a rich source of energy: they provide twice the amount of energy as compared to carbohydrates. They are components of the cell membrane, help in the production and regulation of hormones, and help in the absorption of vitamins A, D, E and K (the fat-soluble vitamins), so a moderate amount of fat is needed for vitamins to be absorbed and work properly. Animal sources of fats include meat, poultry, fish, eggs, animal fat, butter, ghee and oil; plant sources are vegetable oils such as olive oil, palm oil, canola oil and coconut oil. Fats from plant sources are considered good for health, while excessive use of animal fats may cause health problems like heart diseases and obesity. A balanced diet therefore includes only a small amount of fats.

3. What are nutritional deficiency disorders? How can we prevent them?

Nutritional deficiency disorders are disorders caused by an unbalanced diet — a diet that lacks the essential components of

food needed by the human body for normal functioning. Examples from the unit: carbohydrate deficiency causes fatigue, weakness, low weight and constipation; protein deficiency causes kwashiorkor and marasmus; vitamin A deficiency affects vision and causes night blindness; vitamin C deficiency causes bleeding of gums (scurvy); vitamin D deficiency causes weakness of bones (rickets); iron deficiency causes anaemia (lack of red blood cells); calcium deficiency causes weak bones; and iodine deficiency causes goiter (swelling in the neck due to an enlarged thyroid gland). They can be prevented by eating a balanced diet containing all essential nutrients in precise proportions, with plenty of water, and by following the food pyramid and 'My Plate' guidance. (Note: the unit's Table 3.4 as printed pairs 'iron deficiency' with 'weight gain, goiter'; the answers above follow the unit's own MCQ 5, which asks for the cause of goiter and answers iodine, and Table 3.2, which says iron deficiency may cause anaemia.)

4. Correlate balanced diet and fitness keeping in view the age of a person.

Physical and mental health depends on a balanced diet and regular exercise — 'we are what we eat'. The unit's recommended diet chart for people of normal health doing routine work is: children (1-12 years): 0-18 months breast feeding; 18 months onwards fruits, grains, vegetables; 4-6 years fruits, vegetables, meat (chicken, fish), yogurt, grains and milk; 7-12 years fruits, vegetables, whole grains, low fat milk, yogurt. Adults (13-19 years): calcium rich food (dairy products, leafy vegetables), iron rich food, vitamin D rich food (fish, dairy products), sunlight exposure, fruits,

vegetables, meat. Young adults (20-39 years): whole grains, calcium, iron and vitamin D rich products, vegetables, fruits, meat, dry fruits, yogurt. Old age (40-60 years): grains (brown rice, wheat), vegetables (leafy), carrot, fruits, lean protein (chicken, fish), low fat milk (yogurt, cheese), nuts, vitamin D, calcium and iron rich food. So as age changes, the emphasis shifts from growth foods (milk, grains, fruits) in childhood to calcium, iron and vitamin D in adolescence and adulthood, and to lean protein, low fat milk and nuts in old age, always with regular exercise for fitness.

5. What are dietary minerals? What disorders are caused by the deficiency of minerals?

Dietary minerals are individual elements like calcium, phosphorus, magnesium, potassium, iron, sodium, iodine and zinc, required in small quantities for maintaining strong bones and teeth, regulating body fluids, supporting the immune system and strengthening muscle contraction. They are found in milk, meat, cereals, vegetables, fruits, egg and fish. Disorders caused by mineral deficiency: calcium deficiency causes weak bones (calcium also makes bones and teeth strong, strengthens muscles and helps blood clotting); iron deficiency causes anaemia, or lack of red blood cells (iron is part of haemoglobin); iodine deficiency causes goiter (swelling in the neck due to an enlarged thyroid gland).

Exercise 4.4: Match column A with column B (page 67)

Carbohydrates are composed of -> ?

Carbohydrates are composed of Carbon, hydrogen, oxygen. (Carbohydrates are made up of carbon, hydrogen and oxygen;

glucose is an example.)

Proteins are made up of -> ?

Proteins are made up of Amino acids. (Proteins are large molecules made up of amino acids; there are about 20 different types.)

Fats provide more energy than -> ?

Fats provide more energy than Carbohydrates. (Fats provide twice the amount of energy as compared to carbohydrates.)

A conical shaped diagram -> ?

A conical shaped diagram -> Food pyramid chart. (The food pyramid is a guide chart, a conical shaped chart: its base shows the maximum amount of food we can take, its top the food we should take minimum — fats, oils and confectionery sparingly; meat, fish and alternatives 2 small servings; milk, cheese and dairy 3 servings; fruit and vegetables 2-4 servings; bread, cereals and potatoes 4 servings.)

Vitamin D deficiency -> ?

Vitamin D deficiency -> Bone weakness. (Vitamin D protects from rickets, in which bones become soft and weak; it is obtained from liver, egg, fish, mushrooms and dairy products, and the skin synthesizes it in sunlight.)

Carbohydrate deficiency cause -> ?

Carbohydrate deficiency cause -> Fatigue. (Carbohydrate deficiency causes fatigue, weakness, low weight and constipation.)

Old age -> ?

Old age -> Lean protein and low fat milk. (The unit's diet chart for old age, 40-60 years, includes lean protein (chicken, fish) and low fat milk (yogurt, cheese), with grains, leafy vegetables, carrot, fruits, nuts, vitamin D, calcium and iron rich food. The extra Column B entry 'Roughage' matches nothing and is left unused.)

Exercise 3.5: Suggest balanced diet for different age groups (pages 67-68)

Fill the table: balanced diet for Children (1-12 years), Adults (13-19 years) and Young adults (20-39 years).

Children (1-12 years): 0-18 months breast feeding; 18 months onwards fruits, grains, vegetables; 4-6 years fruits, vegetables, meat (chicken, fish), yogurt, grains and milk; 7-12 years fruits, vegetables, whole grains, low fat milk, yogurt. Adults (13-19 years): calcium rich food (dairy products, leafy vegetables), iron rich food, vitamin D rich food (fish, dairy products), sunlight exposure, fruits, vegetables, meat. Young adults (20-39 years): whole grains, calcium, iron and vitamin D rich products, vegetables, fruits, meat, dry fruits, yogurt. (Old age, 40-60 years, from the unit's chart: grains (brown rice, wheat), leafy vegetables, carrot, fruits, lean protein (chicken, fish), low fat milk (yogurt, cheese), nuts, vitamin D, calcium and iron rich food.)

Exercise 3.6: Project (page 68)

Project: Investigate the following using library and internet: nutritional deficiency disorders, their causes and prevention; and

suggest a balanced diet for your grand parents. Guidance from the unit: deficiency disorders include kwashiorkor/marasmus (protein), night blindness (vitamin A), scurvy with bleeding gums (vitamin C), rickets and bone weakness (vitamin D, calcium), anaemia (iron) and goiter (iodine); they are prevented by a balanced diet with all nutrients in precise proportions. For grand parents (old age, 40-60 years and above), the unit's chart suggests grains (brown rice, wheat), leafy vegetables, carrot, fruits, lean protein (chicken, fish), low fat milk (yogurt, cheese), nuts, and vitamin D, calcium and iron rich food. Write your investigation and suggestion in your own words.

Unit 4: Human Digestive System

Exercise 4.1: Tick the correct option (pages 78-79)

1. Digestion means the breaking down of food into:

- Large insoluble molecules
- Large soluble molecules
- Smaller insoluble molecules
- ✓ Smaller soluble molecules

2. Which of the following is NOT a part of the human digestive system?

- Stomach
- Small intestine
- Large intestine
- ✓ Kidneys

3. Digestion does not occur in:

- Mouth
- Stomach
- ✓ Esophagus
- Colon

4. Gastric juice secreted by stomach contains:

- Saliva
- Bile
- Lipase

✓ Hydrochloric acid

5. Lipase is secreted by:

Salivary gland

Gastric gland

✓ Pancreas

Liver

6. The enzyme which digests fat is called:

Amylase

Maltase

Protease

✓ Lipase

7. Proteins are digested by the enzyme:

Sucrase

Amylase

Lipase

✓ Protease

8. A digestive disorder is:

Fever

✓ Constipation

Cold

Cough

9. Constipation occurs due to:

Slow peristalsis

Poor diet

Dehydration

✓ All of these

10. Symptoms of ulcer include:

Abdominal pain

Nausea

Bleeding

✓ All of these

Exercise 4.2: Write short answers (pages 79-80)

1. Define the process of digestion.

Digestion is the process by which the body breaks down food into smaller diffusible particles that can be absorbed into blood and utilized for energy, growth and maintenance of the body. Food consists of large pieces which cannot be absorbed, so they must be broken down into smaller, soluble and diffusible molecules.

2. How physical digestion differs from chemical digestion?

Physical digestion is the mechanical breakdown of food into smaller particles, making it easier for enzymes to act on it; it starts in the mouth (teeth break food into small pieces) and continues in the stomach and small intestine. Chemical digestion is the process by which enzymes break down food particles into smaller and simpler components that can be absorbed easily; enzymes are proteins that speed up the reactants into products but themselves remain unchanged.

3. Write names of any three enzymes which help in digestion.

Salivary amylase (in saliva, breaks down carbohydrates into simpler sugars); pepsin (in stomach, breaks down proteins); and lipase (in pancreatic juice, digests fat into fatty acids and glycerol). Other enzymes named in the unit: amylase (starch into maltose), proteases (proteins into amino acids), maltases (carbohydrates into glucose), trypsin (in pancreatic juice).

4. Write names of three parts of small intestine.

The three parts of the small intestine are the duodenum (first part), the jejunum (second part) and the ileum (third part). The small intestine is a long coiled muscular tube between the stomach and large intestine, about 6 metres long, and the longest part of the digestive system.

5. Write names of digestive disorders. (printed as item 5 of 4.2 on page 80)

The common digestive disorders named in the unit are diarrhoea, constipation, ulcers and indigestion.

Exercise 4.3: Write detailed answers (page 80)

1. Draw a neat and labeled diagram of Human digestive system and briefly describe alimentary canal.

The human digestive system consists of the alimentary canal and the digestive glands. The alimentary canal is a long tube, about 30 feet long (the unit also notes about 10 m, with food taking about 24 hours to pass from one end to the other), that starts from the mouth and ends at the anus. Its parts, in order: oral cavity (mouth;

teeth mechanically digest food, tongue with taste buds mixes saliva with food to make a soft bolus; pharynx is a common path for food and air); esophagus (a short, narrow, muscular tube connecting the oral cavity and stomach; food balls are pushed into the stomach by peristalsis, the wave-like muscle contractions; no digestion takes place in it); stomach (a muscular J-shaped sac-like organ in the upper abdomen, with valves at both ends; its wall secretes gastric juice containing hydrochloric acid, which kills unwanted bacteria, and the enzyme pepsin, which digests protein; food remains inside for 1 to 6 hours); small intestine (a long coiled muscular tube, about 6 metres, divided into duodenum, jejunum and ileum; bile from the gall bladder and pancreatic juice from the pancreas are poured into the duodenum, and the enzymes in pancreatic juice complete digestion; the jejunum absorbs digested diffusible nutrients through finger-like projections called villi; the ileum also absorbs nutrients and passes undigested waste to the large intestine); large intestine, also known as colon (terminal part, about 1.5 m; its walls absorb water and minerals into the blood; remaining waste moves into the rectum and is stored there for 12 to 24 hours; muscles in the rectum wall control the release of faeces through the anus — egestion or defecation). Label the diagram with: salivary glands, mouth, pharynx, esophagus, liver, gall bladder, stomach, pancreas, small intestine, large intestine, appendix, rectum and anus.

2. Describe the role of esophagus and stomach in digestive system.

Esophagus: a short, small, narrow muscular tube that connects the oral cavity and stomach. Balls of food are pushed into the stomach

by peristalsis (wave-like muscle contractions). No digestion takes place in the esophagus; it is just a passage way. Stomach: a muscular J-shaped sac-like organ in the upper abdomen, with valves at both ends to control food entry and exit. Its wall secretes gastric juice, which contains hydrochloric acid (HCl) and the enzyme pepsin: HCl kills unwanted bacteria and pepsin digests protein into smaller pieces. Food remains inside the stomach for 1 to 6 hours depending upon the nutrients of the food taken, and the stomach mixes and churns food with stomach juice, breaking it down into small particles.

3. Briefly describe diarrhea and constipation.

Diarrhoea: passing loose stool more than 3 times a day, characterized by abdominal cramps, bloating, nausea and sometimes vomiting. It usually occurs due to irritation of the digestive tract by foreign agents (virus, bacteria) or due to any medicine or rotten food; irritation causes faster and more frequent movements, so water and nutrients are not absorbed and are taken away in faeces. It causes dehydration, so the patient must take plenty of boiled water with minerals like oral rehydration salt (ORS), and fresh fruits and properly cooked vegetables. Constipation: the opposite of diarrhoea — infrequent bowel movements with dry, hard stool and difficulty in passing out. It occurs due to slow peristalsis because of poor diet, dehydration or physical inactivity; due to slow peristalsis more water is absorbed by the wall of the large intestine and the stool becomes dry and hard. The patient should take plenty of water, food rich in fibers like fruits and vegetables, and physical activity and exercise also relieve

constipation.

4. What are enzymes? How do they help in the process of digestion?

Enzymes are biological molecules, protein in nature. They speed up the chemical reactions in the body: digestive enzymes help in the breakdown of complex molecules into smaller ones, while themselves remaining unchanged. For example: amylases digest starch into maltose; lipases digest fat into fatty acids and glycerol; proteases digest proteins into amino acids; maltases break down carbohydrates into glucose. In digestion: saliva contains salivary amylase, which breaks down carbohydrates into simpler sugars; the stomach contains pepsin, which breaks down proteins; and pancreatic juice is rich in enzymes like amylase, lipase and trypsin, which complete the digestion of food in the small intestine.

5. Evaluate that how small intestine help in absorption of nutrients during digestion?

The small intestine is the most important absorbing organ of the alimentary canal: about 90% of nutrient absorption takes place in the small intestine. Its wall contains small finger-like projections called villi, which contain blood vessels. Digested, diffusible nutrients are absorbed through the villi into the blood, so the food becomes part of the blood and is then transported by blood to different parts of the body. Bile from the gall bladder and pancreatic juice from the pancreas are poured into the duodenum (first part), where the enzymes complete digestion; the jejunum (second part) absorbs the nutrients through the villi; and the ileum (third part)

also absorbs nutrients and passes the undigested waste to the large intestine.

Exercise 4.4: Match column A with column B (page 80)

Chemical digestion -> ?

Chemical digestion -> Enzymatic breakdown of food particles. (Chemical digestion is the process by which enzymes break down food particles into smaller and simpler components.)

Alimentary canal -> ?

Alimentary canal -> It is about 30 feet long. (A long tube that starts from the mouth and ends at the anus.)

Physical digestion -> ?

Physical digestion -> Mechanical break down of food. (Physical digestion is the mechanical breakdown of food into smaller particles.)

Teeth are present in -> ?

Teeth are present in -> Oral cavity. (In the oral cavity, food is mechanically digested by the teeth.)

Duodenum -> ?

Duodenum -> The first part of small intestine. (Bile and pancreatic juice are poured into it.)

Villi -> ?

Villi -> Finger-like projections. (Small finger-like projections on the

intestinal wall, containing blood vessels, through which nutrients are absorbed.)

J-shaped sac-like organ -> ?
J-shaped sac-like organ -> stomach. (The stomach is a muscular J-shaped sac-like organ in the upper abdomen.)

Gall bladder -> ?
Gall bladder -> Bile is stored in. (Bile is produced in the liver and stored in the gall bladder. The extra Column B entry 'Liver' matches nothing as a pair and is left unused.)

Exercise 4.5: Name the organ or part for each job (page 81)
For each job, name the organ or part: chews food; swallows food; releases gastric juice; produces bile; absorbs nutrients; releases saliva; churns and grinds food; absorbs water and salts; produces enzymes; stores faeces.
Chews food: Teeth (in the oral cavity/mouth). Swallows food: Pharynx (the common path for food and air; the soft bolus is swallowed by the pharynx into the esophagus). Releases gastric juice: Stomach (its wall secretes gastric juice containing hydrochloric acid and pepsin). Produces bile: Liver (bile is produced in the liver and stored in the gall bladder). Absorbs nutrients: Small intestine (jejunum and ileum, through the villi; about 90% of nutrient absorption). Releases saliva: Salivary glands (saliva contains salivary amylase). Churns and grinds food: Stomach (it mixes and churns food with stomach juice). Absorbs water and salts: Large intestine (colon; its walls absorb water and minerals

into the blood). Produces enzymes: Pancreas (pancreatic juice rich in amylase, lipase, trypsin; the stomach and salivary glands also produce enzymes). Stores faeces: Rectum (remaining waste is stored there for 12 to 24 hours before release through the anus).

Exercise 4.6: Project (page 81)

Project: Investigate the causes of constipation and its complications if it persists for a long time, and suggest preventive measures against constipation. Guidance from the unit: constipation is infrequent bowel movements with dry, hard stool and difficulty in passing out, caused by slow peristalsis because of poor diet, dehydration or physical inactivity; due to slow peristalsis more water is absorbed by the large intestine wall and the stool becomes dry and hard. Prevention: plenty of water, food rich in fibers like fruits and vegetables, and physical activity and exercise. For complications of long-persisting constipation and further detail, investigate with your teacher's help and write in your own words; the unit does not list complications.

Unit 5: Matter as Particles

Exercise 5.1: Tick the correct option (pages 96-97)

1. According to particle state of matter, the particles are in continuous state of:

- ✓ Motion
- Motionless
- Repulsion
- Condensation

2. Particles in solids are:

- ✓ Tightly packed
- Loosely packed
- Freely moving
- Randomly moving

3. Which of the statement is true about liquids?

- ✓ Can flow easily
- cannot flow
- Have definite shape
- Definite volume

4. The particles of gases are:

- Motion less
- Tightly packed
- ✓ Collide with each other

Fixed shape

5. The movement of particles from higher concentration to lower concentration is called:

Inflation

Transfusion

✓ Diffusion

Emission

6. The conversion of solid into liquid at specific temperature is called:

Evaporation

Transpiration

✓ Melting

Boiling

7. Water boils at:

80°C

90°C

✓ 100°C

110°C

8. The process of conversion of solid state directly into gaseous state is called:

Melting

Boiling

Condensation

✓ Sublimation

9. The process of conversion of gas into liquid state upon cooling is known as:

- Freezing
- Sublimation
- ✓ Condensation
- Evaporation

10. The rate of evaporation depends upon:

- Particle size
- Temperature
- Movement of particles
- ✓ All of these.

Note on Q3: two of its options are true for liquids — (a) 'Can flow easily' and (d) 'Definite volume' (the unit says liquids do not have a definite shape but do have a definite volume). (a) is ticked as the intended answer; (d) is also correct by the unit's own text.

Exercise 5.2: Write short answer (page 97)

1. What is particle theory of matter?

Particle theory of matter is the model scientists developed to explain the particle nature of matter. It states: all matter is composed of tiny particles called atoms and molecules; the particles are in continuous motion; there is space between the particles; the particles of matter have force of attraction among them; and heat affects the movement of particles — heat makes the particles move faster.

2. How do particles in solids differ from the particles in liquid?

In solids, particles are tightly packed with minimal space among them, arranged in a regular, three-dimensional pattern called a crystal lattice, locked in their positions with only vibration and a strong force of attraction — so solids do not flow and have a definite shape and volume. In liquids, particles are close to one another but not orderly arranged; they are free to move, so the liquid can flow and take the shape of its container, while the force of attraction among them gives liquids a definite volume. Examples of solids: iron, copper, sugar, salt. Examples of liquids: water, oil, vinegar, mercury.

3. How do you convert liquid into solid?

By cooling the liquid. Freezing is the conversion of liquid into solid upon cooling at a specific temperature, called the freezing point: when liquids are cooled down, the particles move slowly and get fixed positions in a particular shape and convert into solid. The freezing point of water is 0°C.

4. Differentiate between boiling and evaporation.

Boiling is the process of conversion of a liquid into a gas by heating at a specific temperature (the boiling point); when a liquid is heated, the particles gain energy and move faster, so the attractive forces weaken and the liquid changes into gas. The boiling point of water is 100°C. Evaporation is the process of escaping of liquid particles (e.g. water) from the surface of an open container; it takes place at all temperatures, as particles of a liquid evaporate after absorbing heat from the atmosphere, and its rate increases when

the temperature is increased (water evaporates faster in hot summer than in cold winter). So boiling is a bulk change at a fixed temperature, while evaporation is a surface change at all temperatures.

5. What are the freezing and boiling points of water?

The freezing point of water is 0°C and the boiling point of water is 100°C. (The melting point of water is also 0°C.)

Exercise 5.3: Write detailed answers (pages 97-98)

1. Discuss the position and arrangement of particles in three states of matter.

Solids: particles are tightly packed with minimal space among them, arranged in a regular, three-dimensional pattern called a crystal lattice. Particles are locked in their positions with minimal movement — they only vibrate in their place and do not change position — and there is a strong force of attraction among them, so solids do not flow over a surface and have a definite shape and volume. Liquids: particles are close to one another but not orderly arranged; they are free to move, allowing the liquid to flow and take the shape of the container in which they are poured; they are not as tightly packed as in a solid, so liquids do not have a definite shape, but the force of attraction among them gives them a definite volume; particles move in random directions with no fixed position or orientation. Gases: particles have large empty spaces and are free to move in any direction, so gases have no fixed shape and volume; when a gas is put in a container it quickly takes up any available space; gas particles have higher kinetic energy, so they

collide with one another and with the walls of the container; the force of attraction between gas particles is less than in both solids and liquids.

2. Write down two evidences in favour of particle theory of matter.

Two evidences from the unit: (1) Adding air to expand a basketball/balloon — when you inflate a balloon it always goes down after a while, because the balloon has millions of tiny holes through which the small air particles escape and the balloon deflates. (2) Compressing air in a syringe — take a clean empty syringe without a needle, put your finger on the open side and press the plunger; it becomes harder to press because of the air pressure inside the syringe. (Other evidences in the unit: dissolving sugar in water — the sugar dissolves and no sugar particles are visible because particles of sugar move into the empty spaces of water molecules; evaporating salt water — on boiling, the water molecules convert into steam (gas) and salt is left behind.)

3. Give a comparison of three states of matter, in the light of particle theory of matter.

Solids: particles tightly packed, fixed positions, only vibrate, strong force of attraction, definite shape and volume, do not flow (iron, copper, sugar, salt). Liquids: particles close but not orderly arranged, free to move, flow easily, take the shape of the container, definite volume but no definite shape (water, oil, vinegar, mercury — mercury is the only metal on Earth that is liquid at room temperature). Gases: particles widely spaced with large empty spaces, free to move in any direction, collide with one another and

the container walls, no fixed shape and volume, take up any available space, higher kinetic energy, weakest force of attraction (air, hydrogen, oxygen, nitrogen, helium). In short: solids cannot flow, liquids and gases can flow easily; gases and liquids take the shape of their container but solids do not; solids have the strongest attraction and gases the weakest.

4. How sublimation differs from condensation?

Sublimation is the process of conversion of a solid directly into the gaseous state without undergoing the intermediate liquid state — for example iodine, ammonium chloride and naphthalene, and dry ice (solid carbon dioxide directly into carbon dioxide gas; dry ice is used to keep materials cold and dry). Condensation is the opposite direction and a different change: the process of conversion of gas into liquid state upon cooling — when a gas is cooled down, the particles come close together and change into the liquid state (for example, moisture in warm air condensing on a cold glass).

5. Write down main points of particle theory of matter.

The main points of particle theory of matter: (1) All the matter is composed of tiny particles called atoms and molecules. (2) The particles are in continuous motion. (3) There is a space between the particles. (4) The particles of matter have force of attraction among them. (5) Heat affects the movement of particles — heat makes the particles move faster.

6. Apply the particle theory of matter to explain diffusion.

Diffusion is the movement of particles from the area of higher

concentration to the area of lower concentration. By particle theory: particles are in continuous motion, and gas particles move faster than liquid particles, so diffusion takes place in liquids and gases but not in solids, as particles of a solid cannot move from their place; gases show more diffusion than liquids because movement of particles in gases is faster than in liquids. Large and heavy particles show a lower rate of diffusion than small and light particles, and an increase in temperature makes particles move faster, so the rate of diffusion also increases. Example from the unit: a few drops of ink in a beaker of water first form a small dark droplet, then gradually spread and diffuse, and after about 1 hour the entire solution becomes a uniform pale colour.

Exercise 5.4: Match column A with column B (page 98)

Matter exists in -> ?

Matter exists in -> Three physical states. (Matter exists in three physical states: solid, liquid and gas.)

Have definite shape and volume -> ?

Have definite shape and volume -> Solids. (Solids have a definite shape and volume because their particles are tightly packed in a crystal lattice.)

Can flow easily -> ?

Can flow easily -> Liquids. (Liquid particles are free to move, so liquids flow and take the shape of the container.)

Particles have large empty spaces in -> ?

Particles have large empty spaces in -> Gases. (Gas particles have large empty spaces and are free to move in any direction.)

Can be directly converted from solid to gaseous state -> ?
Can be directly converted from solid to gaseous state -> Ammonium chloride and dry ice. (Sublimation: ammonium chloride and dry ice change directly from solid to gas without a liquid state. The extra Column B entry '95°C' matches nothing and is left unused.)

Water freezes at -> ?
Water freezes at -> 0°C. (The freezing point of water is 0°C.)

Water boils at -> ?
Water boils at -> 100°C. (The boiling point of water is 100°C.)

Evaporation is fast in -> ?
Evaporation is fast in -> Hot summer. (Water evaporates faster in hot summer than in cold winter; the rate of evaporation increases with temperature.)

Exercise 5.5: Separate into solids, liquids and gases (pages 98-99)
Separate: Iron, Water, Blood, perfume, Steel, Wood, Honey, Salt, Dry ice, Air, Methane, Milk, Oil, Nitrogen, Steam, Mercury, Oxygen.
Solids: Iron, Steel, Wood, Salt, Dry ice. Liquids: Water, Blood, perfume, Honey, Milk, Oil, Mercury. Gases: Air, Methane, Nitrogen, Steam, Oxygen. (Perfume is a liquid; dry ice is solid carbon dioxide; steam is gaseous water; mercury is the only metal liquid at room

temperature.)

Exercise 5.6 (printed 6.6): Project (page 99)

Project: Investigate using library and internet: (1) what factors affect the rate of evaporation; (2) the role of evaporation in the water cycle. Guidance from the unit: evaporation is the escape of liquid particles from the surface of an open container, at all temperatures, after absorbing heat from the atmosphere; its rate increases when temperature increases (water evaporates faster in hot summer than in cold winter). In the water cycle, evaporation of water is followed by condensation and precipitation (see the unit's Figure 5.14). Investigate further factors and the full cycle with your teacher's help and write in your own words; the unit does not list other factors.

Unit 6: Elements and Compounds

Exercise 6.1: Encircle the correct option (pages 117-118)

1. The smallest particle of matter is called:

✓ Atom
Molecule
Compound
Element

2. Protons are charged particles having:

✓ Positive charge
Negative charge
No charge
Semi charge

3. A group of two or more atoms that join chemically is called:

Mixture
✓ Molecule
Compound
Element

4. A molecule consisting of three atoms is called:

Monoatomic molecule
Diatomic molecule
✓ Triatomic molecule
tetra-tomic molecule

5. An ozone molecule is consist of:

- One atom
- Two atoms
- ✓ Three atoms
- Four atoms

6. A water molecule H2O consist of:

- ✓ Two elements
- Three elements
- Two atoms
- Four atoms

7. The total number of elements known so far are:

- ✓ 118
- 120
- 122
- 130

8. The number of naturally occurring elements is:

- 80
- 85
- ✓ 92
- 118

9. The number of proton present in hydrogen is:

- ✓ 1
- 2

3

4

10. The number of protons present in an atom is called:

Proton number

✓ Atomic number

Mass number

Elements number

Note on Q6: a water molecule H2O consists of two elements (hydrogen and oxygen) and three atoms (two hydrogen atoms and one oxygen atom); (a) 'Two elements' is the correct option.

Exercise 6.2: Give short answers (pages 118-119)

1. Define atom, atomic number and mass number.

An atom is the smallest particle of matter that can take part in a chemical reaction and show the chemical properties of the element; it is the building block of matter (the word 'atom' is Greek for 'indivisible'). The atomic number (Z) is the number of protons present in the nucleus of an atom; it is the identity of an element and is unique to each element (hydrogen's atomic number is 1, helium's is 2). The mass number (A) is the total number of protons and neutrons present in the nucleus of an atom, calculated by adding the atomic number (Z) and the number of neutrons (n).

2. How does an atom differ from a molecule?

An atom is the smallest particle of matter that takes part in a chemical reaction; a molecule is a group of two or more atoms that join chemically. Atoms of all elements (except the noble gases) are

unstable and cannot exist independently, so they join together and form molecules to be stable; the atoms of noble gases (helium, neon, argon, etc.) are stable and can exist independently. For example, two oxygen atoms combine to form an oxygen molecule (O_2), while a single atom of helium exists on its own.

3. What is the chemical formula of water?

The chemical formula of water is H_2O: two hydrogen atoms combined with one oxygen atom.

4. What are monoatomic molecules give an example.

Monoatomic molecules are molecules consisting of a single atom. Example: helium (He), neon (Ne) or argon (Ar) — the noble gases exist in pure atomic form in nature. (A molecule of two atoms is diatomic, e.g. O_2 or H_2; a molecule of three atoms is triatomic, e.g. ozone O_3 or water H_2O.)

5. What is periodic table?

The periodic table is the table in which all the known elements have been arranged according to the number of protons (increasing atomic number). There are 118 elements identified so far, of which 92 are naturally occurring and the rest have been synthesized by scientists. Dmitri Mendeleev is the inventor of the modern periodic table.

6. Define compounds.

A compound is a pure substance made up of two or more kinds of elements chemically combined in a fixed ratio. A compound differs

from a molecule: a molecule is formed when two or more atoms of the same element combine (e.g. O2), while a compound is made of two or more different kinds of elements (e.g. water H2O is both a molecule and a compound, since it consists of hydrogen and oxygen in a fixed ratio).

Exercise 6.3: Write detailed answers (page 119)

1. Describe the structure of an atom.

An atom is the smallest unit of matter that retains all the chemical properties of an element. It is composed of two regions: the nucleus, the centre of the atom, which contains protons and neutrons; and the outer region, which holds electrons in orbits around the nucleus. An atom is made up of three fundamental subatomic particles: protons — positively charged particles present in the nucleus (the nucleus is the positively charged centre of the atom); neutrons — subatomic particles present in the nucleus which carry no electrical charge, so they are neutral; electrons — negatively charged particles revolving around the nucleus in different orbits. The number of negatively charged electrons and positively charged protons in an atom are equal, therefore an atom is a neutral particle; the number of neutrons may vary. About 99.9% of the volume of an atom is empty because the electrons orbit so far away from the nucleus (if an atom were the size of a cricket stadium, the nucleus would be the size of a cricket ball). The hydrogen atom is the smallest atom: it has one electron and one proton and no neutron.

2. Differentiate between atom and molecules.

An atom is the smallest particle of matter that can take part in a chemical reaction and show the chemical properties of the element; it is the building block of matter and cannot be broken down into simpler substances by chemical means in the usual sense. A molecule is a group of two or more atoms that join chemically: atoms of some elements like helium, neon and argon can exist in pure atomic form in nature, whereas atoms of elements like hydrogen, oxygen and nitrogen cannot exist as single atoms and always combine with their own atoms or with other atoms. Atoms of all elements except the noble gases are unstable and cannot exist independently, so they join together and form molecules in order to be stable; noble gas atoms are stable and exist independently and do not form molecules. Examples: O2 (two oxygen atoms), H2O (two hydrogen atoms and one oxygen atom), CO2 (one carbon atom and two oxygen atoms), O3 (ozone, three oxygen atoms).

3. Define element. Write down the names and symbols of first 10 elements of periodic table.

An element is the substance made up of only one type of atoms; the atoms of an element contain the same number of protons, and an element is a chemical substance that cannot be broken down into more simpler substances. Examples: hydrogen, oxygen, carbon, nitrogen. Each element is represented by a symbol — a one or two letter abbreviation; the first letter is always capitalized, and if the symbol has two letters the second letter is not capitalized. The first 10 elements of the periodic table: 1. Hydrogen — H (1 proton); 2. Helium — He (2 protons); 3. Lithium — Li (3 protons); 4.

Beryllium — Be (4 protons); 5. Boron — B (5 protons); 6. Carbon — C (6 protons); 7. Nitrogen — N (7 protons); 8. Oxygen — O (8 protons); 9. Fluorine — F (9 protons); 10. Neon — Ne (10 protons).

4. Write down differences between metals and nonmetals.

Metals: hard, solid and heavy; shiny appearance when polished; high boiling point and difficult to melt; good conductors of heat and electricity. Non-metals: usually gases, or solids which melt easily; dull appearance; low boiling point and melt easily; poor conductors of heat and electricity. The elements are classified into three groups on the basis of physical and chemical properties: metals, non-metals and metalloids (metalloids have properties of both metals and non-metals; boron, silicon, arsenic and antimony belong to the metalloid group). Among the first 10 elements of the periodic table, only lithium and beryllium are metals while all others are non-metals, by the unit's classification.

5. Illustrate the formation of compounds like with the help of word equation sodium chloride, methane and glucose.

Word equations from the unit: Sodium + chlorine → Sodium chloride (NaCl, common table salt). Carbon + hydrogen → Methane (CH4). Carbon + hydrogen + oxygen → Glucose (C6H12O6). Nitrogen + hydrogen → Ammonia (NH3). Compounds have very different properties from the elements they are made of: sodium (Na) is a highly reactive metal and chlorine (Cl2) is a toxic gas, yet together they form sodium chloride, harmless and essential for human health; hydrogen is highly flammable and oxygen helps burning, yet water (H2O) formed by their combination helps in extinguishing

fire.

6. Distinguish between element and compound with examples.
Elements: made up of only one type of atoms; cannot be broken down into more simpler substances by chemical reaction; have unique properties; represented by symbols and arranged in the periodic table (examples: H, O, C, N, Na, Cl). Compounds: made up of two or more different elements; can be broken down into constituent elements by chemical reaction; have properties different from the constituent elements; represented by a chemical formula which shows the number and type of elements in the compound (examples: H2O, CO2, NaCl, CH4, NH3).

Exercise 6.4: Match column A with column B (page 119)
Carbon -> ?
Carbon -> 6 protons. (Carbon's atomic number is 6, so a carbon atom has 6 protons.)

Oxygen -> ?
Oxygen -> 8 protons. (Oxygen's atomic number is 8.)

Water -> ?
Water -> H2O. (Two hydrogen atoms and one oxygen atom.)

Ammonia -> ?
Ammonia -> NH3. (Nitrogen + hydrogen, per the unit's word equation.)

Glucose -> ?

Glucose -> C6H12O6. (Carbon + hydrogen + oxygen, per the unit's word equation.)

Graphite -> ?

Graphite -> A form of carbon. (Carbon exists in three forms: coal, graphite and diamond; graphite is a black, soft, slippery solid used in lead pencils.)

Diamond -> ?

Diamond -> Used in Jewelry. (Diamond is a colourless, transparent and hardest solid, used for cutting and drilling equipment — and, as the matching answer, used in jewelry. The extra Column B entry 'H2O2' matches nothing and is left unused.)

Neon -> ?

Neon -> Advertising signs. (Neon is a colourless, odourless, non-reactive noble gas; it glows orange when electricity is passed through it and is used in advertising signs.)

Unit 7: Mixtures

Exercise 7.1: Tick the correct option (pages 131-132)

1. The physical combination of two or more substance is called:

- Atom
- Molecule
- Compound
- ✓ Mixture

2. The liquid part of a solution is called:

- Solute
- ✓ Solvent
- Sediment
- Suspension

3. A mixture having uniform composition and properties is called:

- ✓ Homogeneous
- Heterogeneous
- Uniform
- Non-uniform

4. The process in which mixture is separated by using filter paper is called:

- Sublimation
- Evaporation
- Decantation

✓ Filteration

5. In a solution, the lesser quantity is called:

Solvent

✓ Solute

Insolubility

Solubility

6. An alloy is a mixture which contains:

No element

Only two elements

✓ Two or more elements

None of these

7. Separating two liquids by pouring off the top layer from the mixture is termed:

Evaporation

Distillation

Filtration

✓ Decantation

8. Water is known as universal solvent because:

Nothing dissolves in it

Everything dissolves in it

✓ Most substances dissolve in it

Substances dissolve rarely

9. Which of the following is NOT an alloy:

Steel

Bronze

Red gold

✓ Iron

10. An impure substance is known as:

Element

Atom

Compound

✓ Mixture

Exercise 7.2: Write short answers of the following questions (page 132)

1. What is a mixture?

A mixture is the physical combination of two or more substances in such a way that they can be separated easily by physical means. The substances are gathered physically but not chemically combined, so each keeps its own properties; air, tea, coffee and juice are all examples of mixtures.

2. Is mixture a pure substance?

No, a mixture is not a pure substance; it is an impure substance. Pure substances (elements and compounds) are made of the same type of particles, while mixtures contain different types of particles and may have uniform or non-uniform composition.

3. How mixtures differ from compounds?

In a mixture the substances are only physically combined, so they

keep their own properties and can be separated by physical means, and the composition is not fixed. In a compound the elements are chemically joined in a fixed ratio, the compound has properties different from its elements, and it can only be broken down by chemical means.

4. Differentiate between homogeneous and heterogeneous mixtures?

A homogeneous mixture has uniform composition and properties throughout the sample, for example salt solution, tea or air. A heterogeneous mixture has non-uniform composition and properties, so its parts can be seen separately, for example salad, soil or a burger.

5. What is an alloy? Give examples of alloys.

An alloy is a homogeneous mixture of two or more elements in which one element must be a metal, made by mixing the metals in molten form and letting them cool. Examples are stainless steel (iron with chromium), brass, bronze and red gold (gold with copper).

Exercise 7.3: Write detailed answers of the following questions (page 133)

1. Define and differentiate between elements, compounds and mixtures.

An element is a substance made up of only one kind of atoms, for example hydrogen, oxygen or carbon. A compound is formed when atoms are chemically joined together into molecules of two or more

different elements in a fixed ratio, for example water. A mixture is a physical combination of two or more substances (elements or compounds) that are gathered physically but not chemically combined, so the parts keep their own properties and can be separated by physical means. Elements and compounds are pure substances; mixtures are impure substances.

2. Differentiate between pure substances and mixtures.

Pure substances (elements and compounds) are made of the same type of particles, have uniform composition and consistent properties throughout the sample, and their particles are chemically bonded and cannot be separated by physical means. Mixtures have different types of particles, may be uniform or non-uniform in composition, their individual particles retain their own properties, and they are physically combined so they can be separated by physical means.

3. Describe the process of filtration.

Filtration separates a suspended solid from a liquid. The liquid mixture is poured through filter paper fitted in a funnel: the liquid passes through the filter paper into the beaker below and is called the filtrate, while the solid is trapped on the filter paper and is called the residue. For example, sand particles in water are separated this way, with the water passing through and the sand left on the paper.

4. Explain types of substances that are separated by distillation.

Distillation separates liquids that have different boiling points, since

the boiling points of two liquids cannot be the same. On heating, the liquid with the lower boiling point evaporates first, is condensed and collected separately, while the other liquid is left behind. The unit demonstrates it with salt solution: the water evaporates and is collected, and the salt is left behind in the flask.

5. Demonstrate water as a universal solvent by explaining components of a solution.

Water is called the universal solvent because most solutes dissolve in it and form a solution. A solution is a homogeneous mixture with two components: the solvent, the part present in greater amount (for example water in sugar solution), and the solute, the part present in lesser amount (for example sugar). A solution in which water is the solvent is called an aqueous solution, for example sugar or salt dissolved in water.

6. Investigate different ways to separate mixtures.

Mixtures are separated by filtration (a suspended solid from a liquid, using filter paper), distillation (liquids with different boiling points, by evaporation and condensation), evaporation (a dissolved solid from a liquid, by heating so the liquid evaporates and the pure solid is left behind) and decantation (two liquids that do not mix, by pouring off the top layer, since the less dense liquid floats on the denser one, as oil floats on water).

Exercise 7.4: Separate the mixtures into homogeneous, heterogeneous and alloys (page 133)

Separate into homogeneous, heterogeneous and alloys: Coffee,

Milk, Sea water, Tea, Soil, Steel, Air, Sugar solution, Blood, Burger, Salad, Salt solution, Red gold, Brass, Bronze.
Homogeneous mixtures: Coffee, Milk, Sea water, Tea, Air, Sugar solution, Blood, Salt solution. Heterogeneous mixtures: Soil, Burger, Salad. Alloys: Steel, Red gold, Brass, Bronze. (Classification follows the unit: homogeneous mixtures have uniform composition, heterogeneous mixtures do not, and alloys are homogeneous mixtures of metals.)

Exercise 7.5: Match column A with column B (page 134)
Physical combination of two or more substances -> ?
Physical combination of two or more substances -> Mixture. (A mixture is the physical combination of substances that can be separated by physical means.)

An example of homogeneous mixture -> ?
An example of homogeneous mixture -> Tea. (Tea is a solution with uniform composition throughout.)

An example of heterogeneous mixture -> ?
An example of heterogeneous mixture -> Air. (Note: this is the table's remaining mixture entry; the unit text itself calls air a homogeneous mixture of gases, so this pairing is by elimination and the table is inconsistent here. The extra Column B entry 'Oxygen' matches nothing and is left unused.)

Corrosion resistant material -> ?
Corrosion resistant material -> Steel. (Stainless steel cutlery is

corrosion (rust) resistant and more durable.)

A substance having one kind of atoms -> ?

A substance having one kind of atoms -> Element. (An element is made up of only one kind of atoms.)

Most abundant gas in atmosphere -> ?

Most abundant gas in atmosphere -> Nitrogen. (Air is 78% nitrogen.)

Universal solvent -> ?

Universal solvent -> Water. (Water dissolves most solutes and is therefore called the universal solvent.)

Solution having water as solvent -> ?

Solution having water as solvent -> Aqueous solution. (A solution in which the solvent is water is called an aqueous solution.)

Exercise 7.6: Investigate and suggest methods for separating the following mixtures (page 134)

Suggest methods for separating: Sugar and water; Petrol and water; Different types of cells in a blood sample.

Sugar and water: evaporation — heat the solution so the water evaporates and the pure sugar (dissolved solid) is left behind. Petrol and water: decantation — petrol floats on water because it is less dense, so the top petrol layer is poured off. Different types of cells in a blood sample: this is an investigation task with no single method given in the unit; the unit's methods are filtration,

distillation, evaporation and decantation, and blood-cell separation in a laboratory is done by centrifugation (beyond this unit).

Unit 8: Energy

Exercise 8.1: Encircle the correct option (pages 152-153)

1. Physical quantities are those quantities which can be:

- Performed
- ✓ Measured
- Set
- Preferred

2. The unit used for the measurement of energy is:

- Liter
- Milliliter
- ✓ Joule
- Decibel

3. All living things and the machines need ..... to do work?

- Hard work
- ✓ Energy
- Sound
- Wind

4. Energy obtained from plants is an example of:

- ✓ Renewable energy
- Non-renewable energy
- Kinetic energy
- Potential energy

5. The energy due to motion is called:

- Potential energy
- ✓ Kinetic energy
- Thermal energy
- Chemical energy

6. Solar panel is used to convert:

- Light energy to chemical energy
- ✓ Light energy to electrical energy
- Potential energy to kinetic energy
- Thermal energy to light energy

7. Resource of renewable energy is:

- Oil
- Gas
- Coal
- ✓ Flowing water

8. A non-renewable energy resource is:

- Sun
- Wind
- Flowing water
- ✓ Coal

9. Plants convert solar energy into:

- Potential energy
- Kinetic energy

Sound energy

✓ Chemical energy

10. A non-renewable energy resource is:

Cheaper

✓ Costly

Unlimited

Environmental friendly

Exercise 8.2: Give short answers (pages 153-154)

1. Define energy also mention its unit?

Energy is the ability to do work; all living things and machines need energy for everything they do, from cooking food to running vehicles and glowing a bulb. Energy is a physical quantity, and the unit used for its measurement is the joule.

2. Differentiate between Potential and Kinetic energy.

Potential energy is the energy stored in an object due to its position, for example a stone raised to a height, water stored behind a dam, or a stretched rubber band (elastic potential energy). Kinetic energy is the energy due to the motion of an object, for example a moving ball or flowing water. Potential and kinetic energy are interconvertible: potential energy can change into kinetic energy and back again.

3. What kind of energy is possessed by a stretched rubber band?

A stretched rubber band possesses elastic potential energy, which is the energy stored in an object when it is stretched or

compressed. When the band is released, this stored energy changes into kinetic energy.

4. Why energy is so important in our life?

Energy is important because we need it for everything we do: cooking food, playing, running vehicles, glowing bulbs, cooling and heating our homes and running electronic devices. Plants also need energy for growth and for preparing their food from sunlight.

5. How sound energy is produced? Give examples of sound energy.

Sound energy is produced by vibrations. Examples are the sound of a bell, a drum, a loudspeaker, a car horn, or the sound produced when a bouncing ball hits the ground (part of its kinetic energy changes into sound energy).

6. What are the advantages of renewable energy resources?

Renewable energy resources are cheap and free of cost, environment friendly and produce no pollution, and they have an endless, recoverable supply, unlike non-renewable sources which exist in limited quantity and cannot be replaced once used up.

Exercise 8.3: Write detailed answers of the following questions (page 154)

1. Describe energy as a physical quantity with examples?

Energy is a physical quantity because it can be measured, just like distance, mass or volume. Quantities that can be measured are called physical quantities, each with its own unit: distance in metres or kilometres, mass in grams or kilograms, volume of liquids

in litres or millilitres, and energy in joules. Examples of energy as a physical quantity are the kinetic energy of a moving ball, the potential energy of water stored behind a dam, and the thermal energy of a hot object, all expressed in joules.

2. State law of conservation of energy by giving examples?

The law of conservation of energy states that energy can neither be created nor be destroyed, but it can be changed from one form to another, so the total energy remains constant. For example, in a roller coaster the cars gain potential energy as they climb and change it into kinetic energy as they come down, and back again. In a bouncing ball, gravitational potential energy changes into kinetic energy as it falls, and on hitting the ground part of the kinetic energy changes into elastic potential, sound and heat energy, then back into kinetic and potential energy as it bounces up.

3. Compare renewable and non renewable energy resources?

Renewable sources of energy (sun, flowing water, wind, plants and biogas) are permanent and recoverable, have an endless supply, are cheap and free of cost, and are environment friendly and produce no pollution. Non-renewable sources (coal, oil and natural gas, the fossil fuels) are not permanent and cannot be recovered once depleted: they were formed over millions of years from dead organisms, their reservoirs are limited, they are costly, and burning them causes air pollution and releases carbon dioxide, which raises the earth's temperature (global warming).

4. Make a list of renewable sources of energy, explain solar energy?

Renewable sources of energy are solar energy, flowing water (hydroelectric power), wind energy, energy from plants, and biogas energy. Solar energy is the energy obtained from the sun in the form of heat and light; it is the major source of energy on earth and sustains life, plants use it to make their food, and nowadays solar panels use it to produce electricity for devices such as solar fans and solar bulbs. Solar energy is free of cost, clean and environment friendly, and everlasting and renewable because it is not going to end.

5. What are energy converters? Give an example of energy converter process in our daily life.

Energy converters are devices or processes which convert one form of energy into another form. Examples from daily life: a plant converts light energy into chemical energy stored in food (glucose); a car engine converts chemical energy stored in fuel into heat energy and then kinetic energy that drives the car; a television converts electrical energy into sound and light energy; a washing machine converts electrical energy into kinetic energy and sound energy.

6. Demonstrate and assemble solar panel to operate a small fan.

Demonstration and assembly task (STEAM activity 8.5): connect a small solar panel (10 W) to a small DC fan with wires and a switch, expose the panel to direct sunlight and switch on — the fan rotates, showing that the solar panel converts light energy into electrical energy. There is no single written answer in the book; the working

model is the answer.

7. Design and make a solar water heater.

Design and make task (STEAM activity 8.7): build a solar water heater from copper tubing bent into a serpentine shape and connected in series, fixed on a frame, connected to an insulated water tank and to the water supply, and set at the correct angle to the sunlight. Cold water is heated by the sun in the collector and stored in the insulated tank; test the system for leaks and proper function. There is no single written answer in the book; the working model is the answer.

Exercise 8.4: Match column A with the column B (page 154)

Kinetic Energy -> ?

Kinetic Energy -> Energy due to motion. (Energy due to the motion of an object is kinetic energy.)

Potential Energy -> ?

Potential Energy -> Energy due to position. (Energy stored in an object due to its position is potential energy.)

Flying aeroplane -> ?

Flying aeroplane -> Kinetic energy. (A flying aeroplane is a moving object, so it has kinetic energy.)

Sun -> ?

Sun -> Renewable source of energy. (Solar energy is renewable and everlasting.)

Food -> ?

Food -> Chemical energy. (Food stores chemical energy, which plants make from solar energy.)

Coal -> ?

Coal -> Non-renewable source of energy. (Coal is a fossil fuel and cannot be replaced once used up.)

Television -> ?

Television -> Convert electrical energy into sound and light energy. (A television converts electrical energy into sound and light.)

Washing machine -> ?

Washing machine -> Converts electrical energy into kinetic energy and sound. (A washing machine converts electrical energy into kinetic energy and sound energy.)

Note on Exercise 8.4: the extra Column B entry 'Sound energy' matches nothing and is left unused.

Exercise 8.5: Look at the list of energy converters given below (page 155)

Which converts chemical energy into kinetic energy? (Catapult, Tree, Electric Kettle, Television, A Batsman)

Chemical energy into kinetic energy: A Batsman. (The chemical energy stored in the batsman's food powers the movement of batting.)

Which converts solar energy into chemical energy?

Solar energy into chemical energy: Tree. (A tree converts solar energy into chemical energy stored in food by photosynthesis.)

Which converts elastic potential energy into kinetic energy?

Elastic potential energy into kinetic energy: Catapult. (A stretched catapult stores elastic potential energy, which becomes kinetic energy when released.)

Which converts electrical energy into thermal energy?

Electrical energy into thermal energy: Electric Kettle. (An electric kettle converts electrical energy into heat that boils water.)

Which converts electrical energy into sound and light energy?

Electrical energy into sound and light energy: Television. (A television converts electrical energy into sound and light energy.)

Exercise 8.6: Teacher guide / Project (page 155)

Investigate and suggest measures to reduce energy consumption at home and school.

Investigation task for teacher guidance: investigate and suggest measures to reduce energy consumption at home and school, and explain cheap, renewable and environment-friendly sources of energy (sun, wind, flowing water, biogas) that can be preferred over non-renewable sources. There is no single written answer in the book.

Unit 9: Electricity

Exercise 9.1: Encircle the correct option (pages 168-169)

1. Electrons have:

- Positive charge
- ✓ Negative charge
- Both +ve and -ve charge
- No charge

2. Same charges repel each other while opposite charges:

- Cancel
- Repel
- ✓ Attract
- Subtract

3. The unit used for the measurement of current is:

- Joule
- ✓ Ampere
- Coulomb
- Newton

4. Lightning is an example of:

- ✓ Static electricity
- Current
- Close circuit
- Open circuit

5. An example of conductor is:

- Rubber
- Glass
- Wood
- ✓ Copper

6. An example of insulator is:

- Copper
- Silver
- Aluminum
- ✓ Plastic

7. The path along which electrical current flows is called:

- Hydro circuit
- Magnetic circuit
- ✓ Electrical circuit
- Short circuit

8. A circuit which have only one path for current flow is:

- Parallel
- Closed
- Open
- ✓ Series

9. The circuit having more than one path for current flow:

- ✓ Parallel
- closed

Open

Series

10. The brightness of bulb is NOT affected by:

Length of wire

Thickness of wire

Number of batteries

✓ Shape

Exercise 9.2: Give short answers (page 170)

1. Define charge.

Charge is the property of matter that makes objects attract or repel each other: protons carry a positive charge, electrons carry a negative charge, and neutrons carry no charge. An atom is neutral because it has the same number of protons and electrons; it becomes positively charged when it loses electrons and negatively charged when it gains electrons.

2. Name the device used to measure current.

The device used to measure current is called an ammeter. (The unit of current is the ampere, denoted A.)

3. Differentiate between open and close circuit.

In an open circuit the switch is turned off, so the path is broken and current cannot flow. In a closed circuit the switch is turned on, so the path is complete and current flows through the circuit.

4. How does lightning occur?

Lightning occurs when clouds become electrically charged by rubbing with each other and with the air around them, and clouds with huge opposite charges collide, producing a big spark of light in the atmosphere. Lightning is the major example of static electricity in daily life.

5. Which type of circuit is used in home wiring?

Home wiring uses the parallel circuit, because in a parallel circuit each device has its own path for current, so if one component fails the others continue to function.

Exercise 9.3: Write detailed answers of the following questions (page 170)

1. Define circuit, draw a circuit and write its components.

A circuit is the path along which current flows. A simple electric circuit is a closed path with four components: a source of current (cell or battery), a conductor (copper wire), a load (bulb, fan, etc.) and a switch, which acts as the gateway for the current. Drawing task: draw the circuit with the battery, wire, bulb and switch connected in a closed loop, using the standard symbols (wire as a straight line, cell/battery as long and short parallel lines, bulb as a circle with a cross, switch as a break in the wire).

2. Define and differentiate the series and parallel circuits.

A series circuit has only one path for current flow, with components connected in line one after the other, so the current is the same through the circuit and if one component fails the entire circuit fails. A parallel circuit has more than one path for current flow, with

components connected independently parallel to each other, so the current is different among the components and if one component fails the others continue to function.

3. Describe the characteristic of series and parallel circuits.

Series circuit: only one path for current; components connected in line one after the other; current is the same through the circuit; if one component fails, the entire circuit fails. Parallel circuit: more than one path for current; components connected independently parallel to each other; current is not the same everywhere; if one component fails, the other components continue to function. (Table 9.3 of the unit.)

4. What are the uses of series and parallel circuits in our daily life? Give examples.

Series circuits are used in decoration (fairy) lights, where if one bulb goes out the entire string goes dark; in battery-operated toys, connecting batteries to a motor or LEDs; and in smartphones, connecting batteries, screens and processors. Parallel circuits are used in home wiring and home lighting, in computer hardware (hard drive, RAM and CPU), in cars (headlights, taillights and interior lights), and in power grid stations distributing electricity to homes, businesses and industries.

5. Investigate the factors that affect the brightness of a bulb or the speed of motor.

The factors are: number of batteries — more batteries in a series circuit make the bulb glow brighter and the motor run faster;

number of bulbs — more bulbs in a series circuit share the current, so each bulb becomes dimmer; type of wire — copper wires have high conductivity and low resistance, so the bulb glows brighter; length of wire — longer wires lose more power as heat, so the bulb is dimmer and the motor slower; thickness of wire — thick wires have less resistance and allow more current, so the bulb glows brighter and the motor runs faster.

6. Analyze the role of trip wire in security alarm system.

Trip-wire security alarm task (STEAM activity 9.6): a clothespin holds two screws against the terminals of cells, with a buzzer or LED in the circuit and a thread holding the pin open. When an intruder trips the thread, the clothespin snaps shut, the screws touch, the circuit is completed and the buzzer sounds or the LED lights. The trip wire therefore acts as a switch that closes the alarm circuit when disturbed. (Activity task; the working model is the answer.)

Exercise 9.4: Match column A with the column B (page 170)

Electron -> ?

Electron -> Negative charge. (Electrons carry a negative charge.)

Proton -> ?

Proton -> Positive charge. (Protons carry a positive charge.)

Atom -> ?

Atom -> Neutral particle. (An atom has equal protons and electrons, so it is neutral.)

Load -> ?

Load -> Bulb or a fan. (The load is the appliance that uses the current, like a bulb or a fan.)

Insulator -> ?

Insulator -> Plastic. (Plastic does not allow current to pass, so it is an insulator.)

Attraction -> ?

Attraction -> Opposite charges. (Opposite charges attract each other.)

Electric current -> ?

Electric current -> Flow of electrons. (Current electricity is the flow of charges, i.e. electrons, along a wire.)

Less resistance -> ?

Less resistance -> Thick wire. (Thick wires have less resistance and allow more current to flow.)

Note on Exercise 9.4: the extra Column B entry 'Same charges' matches nothing (same charges repel, they do not attract) and is left unused.

Exercise 9.5: Label the following symbols used in an electric circuit (page 171)

Label the symbol: a straight line.

A straight thick line: Wire. (A wire is the conductor through which

current flows.)

Label the symbol: long and short parallel lines marked + and -.
Long and short parallel lines with + and - terminals: Cell or battery. (A cell or battery is the source of current.)

Label the symbol: a circle with a cross inside.
A circle with a cross inside: Bulb (load). (The bulb uses the current to produce light.)

Label the symbol: a break in the wire with one terminal raised.
A break in the wire with the switch arm raised: Switch (off) / Open circuit. (When the switch is off, the path is broken and no current flows.)

Label the symbol: two terminals joined by a straight line.
Two terminals joined by a straight line: Switch (on) / Closed circuit. (When the switch is on, the path is complete and current flows.)

Exercise 9.6: Teacher guide / Project (page 171)
Investigate the role of a power grid station and the type of circuits used in a grid station.
Investigation task for teacher guidance: investigate the role of a power grid station and the type of circuits used in a grid station for supplying electricity to different sectors of a city. (The unit notes that power grid stations use parallel circuits to distribute electricity.) There is no single written answer in the book.

Unit 10: Magnetism

Exercise 10.1: Tick the correct answer (pages 184-185)

1. The object that attracts the iron or steel is called:

- Cork
- Wood
- Rubber
- ✓ Magnet

2. Like poles of bar magnets:

- Attract
- ✓ Repel
- Behave neutrally
- Spin fast

3. Opposite poles of bar magnets:

- ✓ Attract
- Repel
- Behave neutrally
- Spin slow

4. The magnetic material is:

- Rubber
- Wood
- ✓ Iron
- Plastic

5. The magnets which lose their magnetic property easily are:

✓ Temporary magnet

Permanent magnet

Natural magnet

Normal magnet

6. Hard magnetic material like steel is used to make:

Temporary magnet

✓ Permanent magnet

Natural magnet

Electromagnet

7. The space around the magnet where magnetic material feels a force is called:

Gravitational field

Electrical field

✓ Magnetic field

Magnetic shield

8. A magnetic material can be magnetized by.

Heating

Dropping

Hammering

✓ Electric current

9. Magnetized material can be demagnetized by:

Heating

Dropping

Hammering

✓ All of these

10. When freely moving magnet comes to rest it points in a direction:

East-West

South-West

✓ North-South

North-East

Exercise 10.2: Give the short answers (page 185)

1. Define magnet.

A magnet is an object that produces a magnetic field and attracts magnetic materials such as iron, nickel, cobalt and steel. Natural magnets occur as the mineral magnetite, also called lodestone.

2. What is magnetic field?

The magnetic field is the space around a magnet in which a magnetic material feels a force of attraction or repulsion. It is shown by imaginary lines that originate from the North pole of the magnet and enter into the South pole.

3. What are three types of magnets?

The three types of magnets are temporary magnets, permanent magnets and electromagnets. Temporary magnets keep their magnetism only for a time, permanent magnets do not lose their magnetism easily, and electromagnets produce their magnetic field

from an electric current.

4. Name any three magnetic materials.

Three magnetic materials are iron, nickel and cobalt (steel is also a magnetic material). Materials like rubber, plastic and wood are non-magnetic.

5. What is curie temperature?

Curie temperature is the temperature above which a magnet loses its magnetic properties. Heating a magnetized material above this temperature demagnetizes it.

Exercise 10.3: Write detailed answers of the following questions (page 185)

1. What are magnets? Describe the properties of magnets.

Magnets are objects that produce a magnetic field and attract magnetic materials. Their properties are: attraction and repulsion — a magnet attracts magnetic materials, and like poles repel while unlike poles attract; poles — the magnetic force comes from two points, the North and South poles, where iron filings stick most; pair property — if a magnet is cut into two pieces, each piece is a complete magnet with its own North and South poles; and directive property — a freely moving magnet comes to rest pointing in the North-South direction.

2. What is magnetism? Explain, how a magnetic material can be magnetized?

Magnetism is the force of attraction or repulsion between magnets

and magnetic materials. A magnetic material can be magnetized by rubbing, stroking and by the contact method: for example, rub an iron bar continuously from one end with a bar magnet, keeping the magnet in the same position, and after some time the iron bar becomes magnetized, with one side as the North pole and the other as the South pole, so it can attract small iron pieces. (The exercise's option (d) 'Electric current' is also a way to magnetize, through an electromagnet.)

3. Investigate how a magnetized material can be demagnetized?
Demagnetization is the process by which a magnetized material loses its magnetic property. It can be done by heating the material above its Curie temperature (thermal demagnetization), by applying mechanical stress such as dropping or hammering it, by applying electric current, or by keeping a weak magnet in the magnetic field of a strong magnet.

4. Differentiate between temporary and permanent magnets.
A temporary magnet keeps its magnetism only for a time period and can be demagnetized easily; it is made of soft metals such as iron, for example an iron bar magnetized by rubbing with a permanent magnet. A permanent magnet does not lose its magnetic properties easily and its poles cannot be changed; it is made of hard magnetic material such as steel. (Loudspeakers use strong permanent magnets.)

5. Elaborate electromagnets and identify their applications in daily life.

An electromagnet is a magnet in which the magnetic field is produced by electric current: when current flows through a wire wound as a coil around an iron nail, the coil becomes magnetic and attracts iron nails until the switch is opened. Its magnetism can be increased, reduced, stopped or produced by controlling the current, and its poles can be altered with the flow of current. Applications in daily life: electric motors, generators and electric bells, magnetic lifting in scrapyards, MRI machines, maglev trains, loudspeakers, microphones, and the magnetic strips of credit and debit cards.

Exercise 10.4: Match column A with Column B (page 186)

Magnetic material -> ?

Magnetic material -> Iron. (Iron, nickel, cobalt and steel are magnetic materials.)

used for making a permanent magnet -> ?

used for making a permanent magnet -> Steel. (Permanent magnets are made from hard magnetic material such as steel.)

Non-magnetic material -> ?

Non-magnetic material -> A piece of wood. (Wood is not attracted by magnets. Note: 'Rubber' in Column B is also non-magnetic; the table has two non-magnetic entries for one row, so one is left unused as the distractor.)

Hammering causes -> ?

Hammering causes -> Demagnetization. (Hammering, dropping or heating a magnetized material demagnetizes it.)

Strength of electromagnet increases -> ?

Strength of electromagnet increases -> Metal core. (An electromagnet's strength is increased by its metal (iron) core and by the current in the coil; the metal core is the table's matching entry.)

A device helps in navigation -> ?

A device helps in navigation -> Compass. (A compass needle orients itself towards the Earth's magnetic poles and is used to navigate.)

Magnets were discovered by -> ?

Magnets were discovered by -> Magnes. (The first magnet was found when the shoe nails of the Greek shepherd Magnes stuck to a rock containing magnetite.)

Electrical current also produce -> ?

Electrical current also produce -> Magnetic field. (A magnetic field is produced when an electric current passes through a wire, shown by the deflection of a compass needle.)

Exercise 10.5: Answer the following questions (pages 186-187)

1. Why bar magnet always points to the North-South direction?

A bar magnet always points in the North-South direction because of its directive property (polarity): a freely suspended magnet aligns itself with the Earth's magnetic field, so it comes to rest pointing North-South.

2. In the bar magnet, North Pole corresponds to which pole of Earth's magnetic poles.

The North pole of a bar magnet corresponds to the Earth's magnetic South pole, which is located in the Arctic (near the geographic North pole). It is called the North pole because the North pole of a compass needle points towards it.

3. The South Pole of bar magnet corresponds to which pole of Earth's magnetic pole?

The South pole of a bar magnet corresponds to the Earth's magnetic North pole, which is located near the coast of Antarctica.

4. State whether Earth's magnetic poles and geographic poles are the same.

No, the Earth's magnetic poles and geographic poles are not the same. For example, the Earth's magnetic North pole is about 500 kilometres away from the geographic North pole, and the magnetic pole in the Arctic is actually the South pole of the Earth's magnetic field.

Exercise 10.6: Project / Teacher guide (page 187)

Investigate the uses of magnets in devices and home appliances.

Investigation task for teacher guidance: investigate the uses of magnets in devices and home appliances in which magnets are used, for example loudspeakers, electric bells, generators, motors, MRI machines, maglev trains and compasses. There is no single written answer in the book.

Unit 11: Technology in Everyday Life

Exercise 11.1: Tick the correct option (page 198)

1. The practical application of scientific knowledge is called:

- Science
- ✓ Technology
- Understanding
- Wisdom.

2. Scientific knowledge is NOT obtained through:

- Hypothesis
- Observation
- Experiment
- ✓ Unobservance

3. The practice of growing vegetables, fruits and flower is called:

- Agriculture
- Livestock
- ✓ Horticulture
- Floriculture

4. By growing seasonal plants we get:

- Nitrogen
- ✓ Healthy environment
- Carbon dioxide
- Waste

5. Summer season vegetables are grown in:

- June-July
- August-September
- November-December
- ✓ February-march.

6. Winter season vegetables are grown in.

- April- May
- June-July
- ✓ September-October
- November-December

7. Use of organic fertilizer increases:

- Sown seeds
- Seedlings
- ✓ Growth and yield
- Germination

8. The plant grown in summer is:

- ✓ Marigold
- Egg plant
- Camellia
- Current

Notes on Exercise 11.1: Q3 is mislabelled in print — the second option row is printed (a) Horticulture and (d) Floriculture; read in order, the options are (a) Agriculture, (b) Livestock, (c) Horticulture,

(d) Floriculture. Q8 has TWO defensible answers in the unit: marigold is a summer ornamental plant and eggplant is a summer vegetable; (a) Marigold is ticked as the intended answer and the ambiguity is stated here.

Exercise 11.2: Write short answers of the following questions (page 199)

1. Write names of any five summer season vegetables.

Five summer season vegetables (examples from the unit, Figure 11.1): eggplant, cucumber, okra (ladyfinger) and gourd, with any other summer vegetable such as tomato. (The unit names eggplant in the text and shows eggplant, cucumber, okra and gourd in its summer-vegetables figure; summer vegetables are grown in February-March and harvested in September-October.)

2. Write names of any five winter season vegetables.

Five winter season vegetables (from the unit): spinach, cabbage, turnip, onions, cauliflower. (Others named in the unit are carrots, radish and garlic; winter vegetables are sown in September-October and harvested in February-March.)

3. Differentiate between science and technology.

Science is the knowledge obtained through observations and experiments. Technology is the application of scientific knowledge for practical purposes; it has revolutionized every sphere of life, for example in agriculture and in making useful products with microorganisms.

4. Why do we use fertilizer for growing plants? Give two reasons.

We use fertilizer for growing plants because: (1) it supplies the essential nutrients plants need for growth, such as nitrogen, phosphorus and potassium; (2) it enhances soil fertility and increases plant growth and yield. Organic fertilizers also improve soil health and structure and are free from harmful chemicals.

5. Enlist benefits of growing seasonal plants.

Benefits of growing seasonal plants: they provide fresh food (fresh fruits and vegetables), they make our environment healthy, and ornamental plants grown with them add beauty and aesthetic value. Seasonal plants are suited to their season, so they grow well with fewer inputs.

Exercise 11.3: Write detailed answers of the following questions (page 199)

1. Describe the method of making yogurt at home and write names of any four milk products?

Method of making yogurt at home: take about one litre of raw milk, heat it with stirring to 82 degrees C, then let it cool to around 45 degrees C. Add a spoonful of yogurt starter culture (containing beneficial bacteria) and stir it thoroughly into the warm milk. Place the mixture in a warm place at about 38-40 degrees C and let it incubate for 10-12 hours, then chill it in the refrigerator; the homemade yogurt is ready. Four milk products: yogurt, cheese, butter and cream. (The unit names yogurt, cheese and butter as examples of milk products; cream is given as a fourth common example.)

2. What are beneficial microorganisms? How do they convert milk into yogurt and cheese?

Beneficial microorganisms are tiny living things, like bacteria and yeast, that are useful to mankind and make useful products. They convert milk into yogurt and cheese by fermentation: in yogurt-making, the bacteria of the starter culture ferment the warm milk during incubation; in cheese-making, the milk is curdled (by acid from lemon juice or vinegar and by the enzyme rennet), the whey is separated from the curd, and the curd is shaped into cheese.

3. Write the material and procedure for making cheese at home.

Materials for making cheese at home: 1 litre milk, 1 tablespoon lemon juice, white (distilled) vinegar, salt, cheesecloth and rennet (an enzyme). Procedure: heat the milk at 63 degrees C for 30 minutes to kill harmful bacteria, then let it cool to around 32 degrees C. Add one tablespoon of lemon juice or distilled white vinegar and stir evenly, then let the milk incubate for an hour so the acid curdles it. Add rennet to help coagulate the curds, add salt when the whey has separated from the curd, transfer the curds to cheesecloth and gather its edges to form the cheese into its desired shape.

4. Write on the procedure of making solar oven.

Materials for a solar oven: a cardboard box, aluminum foil, plastic wrap, scissors, dark paper, pencil, glue or tape. Procedure: cut a flap in the top of the box to create the lid, cover the inside of the box with aluminum foil, cover the bottom with black construction

paper, place a small pot or dish in the centre, close the lid with the plastic wrap tight, and cut a small hole in the top of the lid for ventilation. Place the oven in the sun so light rays falling on the aluminum foil reflect into the box. Working: the solar oven converts solar radiation into heat energy — the foil reflects light onto the pot, the dark paper absorbs heat, and the plastic wrap traps heat inside to cook the food.

5. Assemble the components of an electric bell and explain its working principle.

Components of an electric bell: a U-shaped piece of iron wound with wire as an electromagnet coil, a soft iron strip (armature) with an iron hammer on one side and a flexible steel strip on the other, a metal gong, a battery, wire, a switch and an adjustable screw. Assembly: fix the coil on a board, fix the soft iron strip so it stays in contact with the adjustable screw and the hammer can strike the gong, and connect the coil, battery, switch and screw in one circuit. Working: when the button is pushed, current flows through the coil, the electromagnet is magnetized and attracts the soft iron strip, so the hammer strikes the gong and produces sound; the strip then leaves the screw, the circuit breaks, the electromagnet demagnetizes and the strip returns — and the process repeats, so the bell keeps ringing until the button is released.

Exercise 11.4: Match column A with column B (page 199)

A summer Vegetable -> ?

A summer Vegetable -> Eggplant. (Eggplant is the summer vegetable named in the unit.)

A winter vegetable -> ?

A winter vegetable -> Spinach. (Spinach is named among the winter vegetables in the unit.)

Starter culture is used to -> ?

Starter culture is used to -> To turn milk into yogurt. (The yogurt starter culture contains beneficial bacteria which cause fermentation.)

An enzyme used to make cheese -> ?

An enzyme used to make cheese -> Rennet. (Rennet is the enzyme added to milk to coagulate the curds.)

Fertilizers provide -> ?

Fertilizers provide -> Nutrients to plants. (Fertilizers supply essential nutrients like nitrogen, phosphorus and potassium.)

An ornamental plant grown in summer -> ?

An ornamental plant grown in summer -> Daisy. (Daisy is named among the summer ornamental plants in the unit.)

An ornamental plant grown in winter -> ?

An ornamental plant grown in winter -> Jasmine. (Jasmine is named among the winter ornamental flowering plants in the unit.)

Electric bell uses -> ?

Electric bell uses -> Electromagnet. (An electric bell works with an

electromagnet.)

Note on Exercise 11.4: the extra Column B entry 'Amylase' matches nothing and is left unused. (In print, Column A carries the descriptions and Column B the items.)

Exercise 11.5: Make a list of summer and winter season vegetables (page 200)

Make a list of summer and winter season vegetables.

Summer season vegetables: eggplant, cucumber, okra, gourd (from the unit's Figure 11.1; summer vegetables are grown in February-March and harvested in September-October). Winter season vegetables: spinach, cabbage, turnip, onions, cauliflower, carrots, radish, garlic (from the unit; winter vegetables are sown in September-October and harvested in February-March).

Exercise 11.6: Project (page 200)

Investigate the importance of fertilizers for plants and compare organic fertilizers with chemical fertilizers.

Investigation task: investigate the importance of fertilizers for plants and compare organic fertilizers with chemical fertilizers. Organic fertilizers (from animal waste, bones and plant matter, including green manure) are rich in nitrogen, phosphorus and potassium, are free from harmful chemicals and promote soil health and structure. The unit's Activity 11.1 compares pots with soil only, soil with farmyard manure and soil with chemical fertilizer by germination and seedling growth. There is no single written answer in the book.

Unit 12: Solar System

Exercise 12.1: Tick the correct option (pages 211-212)

1. The number of planets in our solar system are:

- 6
- ✓ 8
- 10
- 12

2. The closest planet to the Sun is:

- Earth
- Venus
- ✓ Mercury
- Neptune

3. The farthest planet from the Sun is:

- Mars
- Jupiter
- Uranus
- ✓ Neptune

4. The dwarf planet is:

- Jupiter
- Neptune
- Earth
- ✓ Pluto

5. Our solar system is part of a galaxy called:

- Andromeda
- Asteroids
- ✓ Milky Way
- Alpha Centauri

6. The celestial body which revolves around a planet is called:

- Dwarf planet
- Comet
- Asteroid
- ✓ Satellite

7. The Sun is composed of:

- Helium and Oxygen
- Oxygen and Nitrogen
- Nitrogen and Hydrogen
- ✓ Hydrogen and helium

8. The hottest planet of the solar system is:

- Earth
- Mercury
- ✓ Venus
- Saturn

9. Halley's Comet is seen after every:

- 56 years
- 66 years

✓ 76 years

86 years

10. GPS is used for:

Space research

✓ Navigation

Cultivator

Weather forecasting

Exercise 12.2: Write short answers of the following questions (page 213)

1. What is solar system?

The solar system is the part of the Milky Way galaxy that includes the Sun and everything revolving around it: the eight planets, comets, asteroids, meteorites and satellites. The Sun is at the centre of the solar system.

2. Differentiate between star and a planet?

A star is a celestial body that emits its own light and energy due to nuclear reactions inside it, for example the Sun. A planet is a celestial body that does not have its own light and energy; planets revolve around the Sun, for example Earth, Mars and Jupiter.

3. Write the names of any three dwarf planets?

Any three dwarf planets: Pluto, Eris and Ceres. (The five officially recognized dwarf planets are Pluto, Eris, Ceres, Haumea and Makemake.)

4. Define an asteroid and write any two characteristics?

An asteroid is a piece of rock that revolves around the Sun. Two characteristics: asteroids are found between Mars and Jupiter, where they form a belt called the asteroid belt; and they vary in size, some being small and some as large as dwarf planets, and they have no atmosphere.

5. How asteroids differ from meteorites?

Asteroids are pieces of rock that revolve around the Sun, mostly in the asteroid belt between Mars and Jupiter. Meteorites are rocks and pieces of rock that originate in outer space (formed by the collision of asteroids) and fall on the surface of a moon or a planet; when a meteorite enters the atmosphere it becomes a meteor and forms a fireball. Meteorites are usually smaller in size and light in weight.

Exercise 12.3: Write detailed answers of the following questions (page 213)

1. Define satellites. Describe different types of satellites.

A satellite is an object that revolves around a planet in a specified orbit. Satellites are of two types. Natural satellites are celestial bodies that revolve around a planet and do not have their own light; the moons of the planets are natural satellites, for example the Moon is the satellite of Earth. Artificial satellites are launched from Earth into different orbits for different purposes: geostationary satellites (35,800 km above the equator, moving west to east, appearing stationary) are used for telecommunication, broadcasting and navigation; weather satellites in low polar orbit

scan the Earth for meteorological conditions and help weather forecasting; communication satellites in geostationary orbit transmit radio, TV and mobile phone signals; and GPS satellites (more than 30 orbiting the Earth) tell the position of something or someone on Earth.

2. Describe characteristics of Mercury and Mars.

Mercury is the closest planet to the Sun, has a rocky surface and no atmosphere, no water is found on it, and it is the smallest planet of the solar system with a diameter of about 4879 km; it becomes extremely hot during the day and cold at night. Mars is also called the red planet due to its red colour; it has an atmosphere and a rocky surface, has frozen water, and scientists are exploring it by sending space missions.

3. What is a star? Explain Sun as a centre of our solar system.

A star is a celestial body that emits its own light and energy due to nuclear reactions inside it. The Sun is a star and the centre of our solar system: it is a medium-sized star, a huge ball of hot and glowing gases (about 75% hydrogen and 25% helium by mass), emitting heat and light continuously, and it is the only source of heat, light and energy for all the planets. The eight planets revolve around the Sun at different distances, and the Earth revolves anticlockwise around the Sun at about 109,000 km per hour.

4. Define a comet and describe Halley's Comet sighting cycle in detail.

A comet is a combination of dust, rock and ice that orbits the Sun in

an elliptical orbit; its size varies from a few miles to tens of miles, and when it comes near the Sun the dust and gases burn and spew up, forming a glowing head and tail. Halley's Comet is the most well-known comet, named after the English astronomer Edmond Halley, who first predicted its return in 1758. It is made of ice, dust and rocky particles, can be seen with the naked eye from Earth every 75-76 years, was last seen in 1986 and will return again in 2061.

5. Investigate how artificial satellites have contributed in space research?

Artificial satellites have improved our knowledge of space and are used for space research in several ways: geostationary satellites support telecommunication, broadcasting and navigation; weather satellites scan the Earth and give the data for weather forecasting; communication satellites transmit signals across the world; and GPS satellites (more than 30) give exact positions on Earth. Together they let scientists observe the Earth and the universe continuously from space, track weather and the environment, and guide space missions — the project in Exercise 12.6 extends this to the Hubble and James Webb space telescopes. (Investigation answer, based on the unit's satellite types.)

Exercise 12.4: Match column A with column B (page 213)

A celestial body having its own light -> ?

A celestial body having its own light -> Star. (Stars emit their own light and energy.)

Geostationary satellites move from -> ?

Geostationary satellites move from -> West to east. (A geostationary satellite revolves in the same direction as the Earth, west to east.)

A dwarf planet -> ?

A dwarf planet -> Makemake. (Makemake is one of the five officially recognized dwarf planets.)

Red planet -> ?

Red planet -> Mars. (Mars is called the red planet due to its red colour.)

Natural satellite -> ?

Natural satellite -> Moon. (The Moon is the natural satellite of Earth.)

Planet on which life exists -> ?

Planet on which life exists -> Earth. (Earth is the only planet in our solar system on which life exists.)

Meteorite -> ?

Meteorite -> Shooting star. (When a meteorite enters the atmosphere it becomes a meteor, seen as a fireball or shooting star.)

First artificial satellite -> ?

First artificial satellite -> Sputnik. (Sputnik was the first artificial

satellite.)

Note on Exercise 12.4: the extra Column B entry 'Sun' matches nothing and is left unused.

Exercise 12.5: Label the diagram of the planets in the solar system (page 214)

Label the planets in order, starting from the planet closest to the Sun.

Starting from the planet closest to the Sun, the order is: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune. (Mercury is closest to the Sun and Neptune is the eighth, last and farthest planet from the Sun.)

Exercise 12.6: Project - Investigate the following (page 214)

Investigate: artificial satellites and space knowledge; Hubble discoveries; James Webb telescope purpose.

Investigation task: (1) How artificial satellites have improved our knowledge about space — see Exercise 12.3 Q5 (telecommunication, broadcasting, navigation, weather forecasting, GPS positioning and continuous observation from space). (2) Important discoveries made with the help of the Hubble space telescope, and (3) the purpose of launching the James Webb telescope and how it will help astronomers understand the universe. The unit names these telescopes only in the project task and gives no content about them, so these are open investigation topics for the student, not answers fixed by the book.